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RESEARCH ARTICLE

LLM-Empowered Resource Allocation in Wireless Communications Systems

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ABSTRACT The recent success of large language models (LLMs) has spurred their application in various fields. In particular, there have been growing efforts to integrate LLMs into wireless communications systems. Leveraging LLMs in this domain offers the potential to realize artificial general intelligence (AGI)-enabled wireless networks. In this paper, we investigate the feasibility and characteristics of an LLM-based resource allocation framework. To fundamentally assess the reasoning capability of LLMs, we formulate a representative resource allocation problem involving two transmit pairs, aiming to maximize either energy efficiency (EE) or spectral efficiency (SE). We propose a few-shot learning-based approach that prompts the LLM with channel conditions to infer resource allocation decisions, thereby eliminating the need for explicit model training. To address potential reliability concerns, we further introduce a hybrid strategy that combines the LLM's output with a conventional low-complexity method, serving as a robust fallback. Through simulations, we demonstrate that the proposed framework achieves up to 96% of the optimal performance in EE maximization. Our performance evaluation also yields two key insights. First, specialized models such as CodeLLaMA demonstrate superior performance, likely due to the structural resemblance between programming tasks and resource allocation problems. Second, the proposed framework achieves more substantial gains in EE optimization, which can be viewed as a continuous approximation problem, compared to SE optimization, which is inherently more discrete and binary in nature. After confirming the feasibility of LLM-based resource allocation through extensive simulations, we address key technical challenges, such as latency, optimal architecture, explainability, and training methodology, that must be overcome for practical deployment, paving the way for the next generation of intelligent wireless systems. Furthermore, the proposed approach is adaptable to a wide range of optimization problems beyond wireless communications, making it applicable to various domains that require intelligent decision-making.

INDEX TERMS Large language model (LLM), few-shot learning, optimization framework, resource allocation, wireless communications, artificial intelligence applications.

I. INTRODUCTION

In wireless communication systems, the allocation of resources such as transmit power, bandwidth, or beamforming is of utmost importance because the open nature of the wireless medium causes interference to neighboring nodes [1], [2]. Recently, the number of transmitting nodes has increased, while the required communication objectives have become more diverse and stringent. Modern networks must now simultaneously accommodate services

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with often conflicting demands, such as the massive data rates for enhanced mobile broadband (eMBB) and the millisecond-level latency for ultra-reliable low latency communications (URLLC), transforming the problem into a multifaceted design challenge. At the same time, more complicated system models involving technologies like massive multiple-input and multiple-output (MIMO), reconfigurable intelligent surfaces (RIS), and millimeter-wave (mmWave) communications are being considered in conjunction with stringent constraints, such as extremely low computation time [3]. These factors complicate resource allocation, spurring extensive research in this field.

Since finding the optimal resource allocation strategy is typically formulated as an optimization problem, analytical optimization frameworks such as convex optimization have been widely applied [4], [5], [6]. However, given that the Shannon capacity formula is nonconvex with respect to transmit power and many control variables, such as channel selection, are discrete, finding the optimal resource allocation is challenging. To address these issues, novel mathematical approaches, such as game-theoretic frameworks or convex relaxation techniques, that approximate the problem into a tractable form, have been developed [4], [7], [8], [9], [10]. Despite these advances, most analytical approaches face limitations in efficiently managing large numbers of nodes at low computational times, since they are often based on iterative algorithms whose convergence time is unpredictable and typically too slow for real-time applications [7], [11], [12], [13]. Moreover, they struggle to adapt to dynamically changing wireless environments, as they are typically tailored to specific scenarios, making it difficult for them to adapt to new objectives or system models.

The deep learning (DL)-based resource allocation has received considerable attention to address the above-mentioned challenges [11], [14], [15], [16], [17], [18], [19]. In DL-based schemes, the optimal resource allocation strategy is approximated using specially structured deep neural network (DNN) models. Since DNNs primarily perform simple matrix operations, once properly trained, they can achieve near-optimal performance with significantly low computation times, on the order of a few milliseconds. Despite these advantages, the design and training of DNNs is task-specific, requiring unique structures and training for each resource allocation task. This limits the applicability of the DL-based approach. To improve the adaptability to varying network conditions, new methods such as the use of graph neural networks (GNNs) have been explored [14], [20]. Nonetheless, challenges remain in generalizing these models to new network requirements and scenarios.

Very recently, the large language model (LLM), which is a foundational model built on natural language understanding, has achieved remarkable success across various domains, as exemplified by OpenAI's ChatGPT and Google's Gemini. These models are widely recognized as game changers, marking a paradigm shift in artificial intelligence (AI) from task-specific narrow intelligence to general-purpose reasoning capabilities. This evolution opens up new possibilities for wireless network design and operation, where LLMs can serve as central reasoning engines, potentially enabling significantly higher levels of performance and autonomy [21], [22], [23], [24]. The key innovation of LLMs lies in their ability to process not only numerical data but also high-level, human-like instructions and contextual information. In particular, the multimodal nature of wireless communication environments can be effectively leveraged in LLM-based frameworks. By integrating heterogeneous data sources, such as spectrum occupancy heatmaps, satellite imagery, and real-time camera feeds—an LLM can construct holistic situational awareness, facilitating predictive and proactive control that moves beyond reactive network management. For instance, the authors of [23] proposed a virtual signal large model for few-shot wideband signal detection and recognition, demonstrating the potential of large-scale models for signal-level spectrum analysis.[1]

Thanks to the human-like reasoning and inference capabilities of LLM, efforts have been made to use them to solve mathematical problems, and it has been demonstrated that LLMs can indeed be effective in this domain [25], [26], [27], [28]. Consequently, it is promising to employ the LLMs to address various optimization problems in wireless communication systems, such as finding optimal resource allocation strategies. This LLM-based approach offers a significant advantage over the conventional DL-based approaches, which require designing and training task-specific models that are often brittle and not generalizable. With LLMs, adaptation to new objectives or network topologies can be achieved simply by modifying the natural language prompt, rather than undertaking a costly re-engineering and retraining process. Thus, the LLM can be considered a general-purpose solver, capable of addressing a wide range of optimization problems and making it highly adaptable to dynamic environments.

In this article, we investigate the feasibility of LLM-based resource allocation, positioning LLMs not merely as tools but as potential general-purpose solvers for complex optimization problems in wireless networks. To purely isolate and verify the foundational mathematical reasoning and contextual understanding capabilities of LLMs, we deliberately adopt a simplified yet representative two-transmit-pair system model. This setup allows us to rigorously evaluate the core potential of LLMs without confounding factors arising from overly complex scenarios.

1) **Application of LLM-based Resource Allocation:** We present the basic principles of LLMs and examine the feasibility and benefits of employing LLMs for resource allocation in wireless communication systems.

[1]While such models focus on physical-layer signal detection and classification, our work differs fundamentally in that it leverages LLMs for high-level decision making and resource allocation. Therefore, the proposed framework complements these prior approaches by extending large-model applications to the decision and control layer of wireless systems.

2) **LLM-Based Framework for Resource Allocation:** We design an LLM-based resource allocation framework that maximizes either spectral efficiency (SE) or energy efficiency (EE) by prompting LLMs with channel conditions. This allows the model to infer near-optimal allocation decisions without explicit gradient computation or task-specific training.
3) **Hybrid Optimization Strategy:** To address potential reliability issues, we introduce a hybrid scheme that combines the LLM-based approach with a conventional low-complexity algorithm, ensuring robustness and consistent performance.
4) **Performance Validation:** Extensive simulations demonstrate that the proposed framework achieves near-optimal results, up to 96% of the optimal performance in EE maximization. Furthermore, we provide valuable insights into the effects of pretraining schemes, model size, and the inherent limitations of LLM-based approaches, along with key directions for future research.

This paper is structured as follows. Section II presents the basic principles of LLMs and discusses the feasibility and benefits of applying them to resource allocation. Section III provides the system model, followed by the proposed LLM-based resource allocation framework and hybrid scheme in Section IV. Section V presents the performance evaluation, and Section VI discusses research challenges and concludes the study.

II. BASIC PRINCIPLE OF LLM AND ITS APPLICATION IN RESOURCE ALLOCATION

In this section, we describe the basic principles of LLM. Then, we turn our attention to the feasibility and benefits of using LLMs for resource allocation in wireless communications systems.

A. PRINCIPLE OF LLM

The term foundation model refers to a type of large-scale DNN trained on vast amounts of unlabeled data using self-supervision techniques. The primary goal is not to solve a single problem, but to create a highly generalizable and powerful base model that can be fine-tuned for a wide range of applications. Notably, these models are characterized by their deep architectures and immense scale, with recent state-of-the-art examples such as GPT-4 reportedly comprising over 1 trillion parameters. This massive scale is crucial, as it enables them to capture intricate data relationships and perform complex tasks that require sophisticated logical reasoning and contextual understanding.

An LLM is a representative example and a specialized class of foundation model, pre-trained on vast corpora of text data. It excels at generating coherent text and performing a wide range of language-based tasks in response to prompts, which are typically natural language descriptions of the desired output. At its core, this generative capability is realized by probabilistically predicting the most likely sequence of words, or tokens, following a given prompt, thereby enabling the construction of meaningful and contextually relevant responses.

LLM is primarily based on the Transformer architecture, which has become the mainstream framework for modern language models. The Transformer contains two main innovations: the attention mechanism and positional encoding [30], [31]. The attention mechanism computes the relevance of each word in a sentence relative to other words, allowing the model to capture dependencies between words regardless of their distance in the text. This mechanism is enhanced by multi-head attention, which enables the model to focus on different parts of the input sequence simultaneously, capturing different contextual aspects. In addition, positional encoding is added to the input embeddings to encode information about the position of each word in the sequence, ensuring that the order and relative positioning of words are taken into account. These innovations enable more efficient training of DNNs and significantly improve their ability to form abstract representations of input data, thereby enhancing the performance and capabilities of LLMs in various language processing tasks.

Most well-known LLMs are pre-trained on billions to trillions of tokens, equipping them with a vast general knowledge base. However, adapting this general knowledge to specific tasks remains challenging. Traditional fine-tuning can align a model to a task, but it is often computationally prohibitive for models of such scale. As an alternative, modern approaches leverage in-context learning (ICL), a paradigm that adapts the LLM's behavior without updating any of its parameters [32]. In ICL, a few task-specific examples (i.e., few-shot learning) are embedded directly within the prompt to elicit the model's emergent capacity for analogy-based reasoning. The model analyzes these examples, infers underlying logical or mathematical patterns, and extrapolates these patterns to solve new, analogous problems. The effectiveness of ICL can be further enhanced by advanced prompting strategies that steer the model's reasoning process. Among these, Chain-of-Thought (CoT) prompting is particularly foundational, as it encourages the model to decompose complex problems into sequences of intermediate logical steps rather than producing an immediate final answer [33], [34]. This technique transforms the LLM from a closed-box pattern matcher into a more transparent and deliberate reasoning engine, thereby unlocking its potential for a wide range of structured reasoning tasks well beyond conventional language processing.

Given their significance, leading AI companies and research institutions have developed their own LLMs, fostering a vibrant ecosystem of both proprietary and open-source models [35].

- **GPT (OpenAI):** GPT, or Generative Pre-trained Transformers, are decoder-only Transformer-based language models developed by OpenAI. They remain a

state-of-the-art benchmark for performance. While early versions were open-source, recent models such as GPT-4 and its successors are closed-source and accessible only via APIs. Their key strengths lie in powerful general reasoning capabilities and advanced multimodality, enabled by massive scale and sophisticated alignment techniques such as Reinforcement Learning from Human Feedback (RLHF) [35].
- **Llama (Meta):** LLaMA, or Large Language Model Meta AI, is a collection of foundational language models released by Meta. A notable aspect of LLaMA is that, unlike GPT, it is open-source, and its model weights are publicly available, which has catalyzed a large ecosystem of community-driven fine-tuning and development. Additionally, LLaMA models are relatively compact compared to GPT-based models, with sizes ranging from 7 billion to 405 billion parameters. As a result, LLaMA has been widely adopted in the research community and for developing task-specific LLMs in mission-critical applications where data privacy and model ownership are essential [36].
- **Mistral/Mixtral (Mistral AI):** Mistral AI has gained considerable attention by developing highly efficient yet powerful open-source language models. The Mistral 7B model demonstrated strong performance relative to its size, while the Mixtral 8×7B model brought widespread attention to the Sparse Mixture-of-Experts (SMoE) architecture. In this design, only a subset of the model's parameters is activated for each input, enabling inference-time efficiency comparable to much larger dense models. This sparsity mechanism significantly reduces computational cost, making the model particularly well-suited for real-world deployment scenarios [37].
- **DeepSeek Series (DeepSeek AI):** The DeepSeek series has emerged as a prominent family of open-source models, demonstrating competitive performance against leading proprietary systems across standard benchmarks. Models such as DeepSeek-V2 adopt a sophisticated MoE architecture, with a particular focus on computational efficiency. DeepSeek applies cost-reduction techniques to enhance the training and inference efficiency of large-scale models, improving their accessibility for broader research usage [38].

B. CONVENTIONAL RESOURCE ALLOCATION STRATEGIES

In wireless communication systems, resource allocation involves adjusting parameters such as transmit power to optimize network performance, e.g., maximizing SE, while satisfying system constraints. Finding the optimal strategy is typically formulated as an complex optimization problem, which can be addressed by using either conventional optimization-based approaches or DL-based approaches as follows [7], [15], [16], [17]:

- **Optimization-based approach:** This approach is based on analytical modeling and formal optimization theory, aiming to derive provably optimal or near-optimal solutions for resource allocation problems. However, practical applications present significant challenges. Realistic wireless network scenarios typically lead to non-convex and NP-hard formulations, often due to the inherent structure of the Shannon capacity formula and the presence of integer decision variables, such as those involved in user scheduling and subchannel assignment. To address these difficulties, researchers have proposed a variety of advanced techniques. One common strategy is the convex relaxation, which approximates the original non-convex problem by a tractable convex surrogate, albeit at the cost of potential performance degradation. Another is the successive convex approximation (SCA), which iteratively solves a sequence of convex subproblems, aiming to converge to a locally optimal solution. Additionally, the game-theoretic models have been employed to capture and analyze strategic interactions among distributed network agents. Despite their strong theoretical foundation, these methods face two critical limitations in practice. First, the high computational overhead associated with iterative procedures often renders them impractical for the real-time decision-making demands of modern wireless networks. Second, their performance heavily depends on the accuracy of the underlying system model and deteriorates as the problem dimension grows, making them less scalable in large, dynamic environments.
- **DL-based approach:** This approach marks a transition from online optimization to offline learning, wherein a DNN is trained to approximate the complex mapping between high-dimensional network state information (e.g., channel matrices) and corresponding resource allocation decisions. Its primary advantage lies in the extremely low inference latency, as a trained DNN can produce near-instantaneous outputs once deployed. Several training paradigms have been proposed for DNN-based resource allocation. Among these, supervised learning is the most commonly employed, where the DNN is trained on a dataset consisting of network states paired with optimal resource allocation labels. For example, the work in [39] proposed a DL-based best-beam prediction and overhead reduction scheme, demonstrating the capability of DNNs to approximate optimal solutions with low computational latency. This approach enables stable and predictable learning behavior and has shown strong performance when high-quality labeled data are available. However, generating such labels can be computationally demanding. Unsupervised learning offers an alternative by directly optimizing performance metrics, e.g., SE, without requiring ground-truth labels, as demonstrated in [40]. While promising, its convergence behavior is more difficult to analyze and may be sensitive to initialization or network structure. Reinforcement learning (RL) frames the problem as an interaction

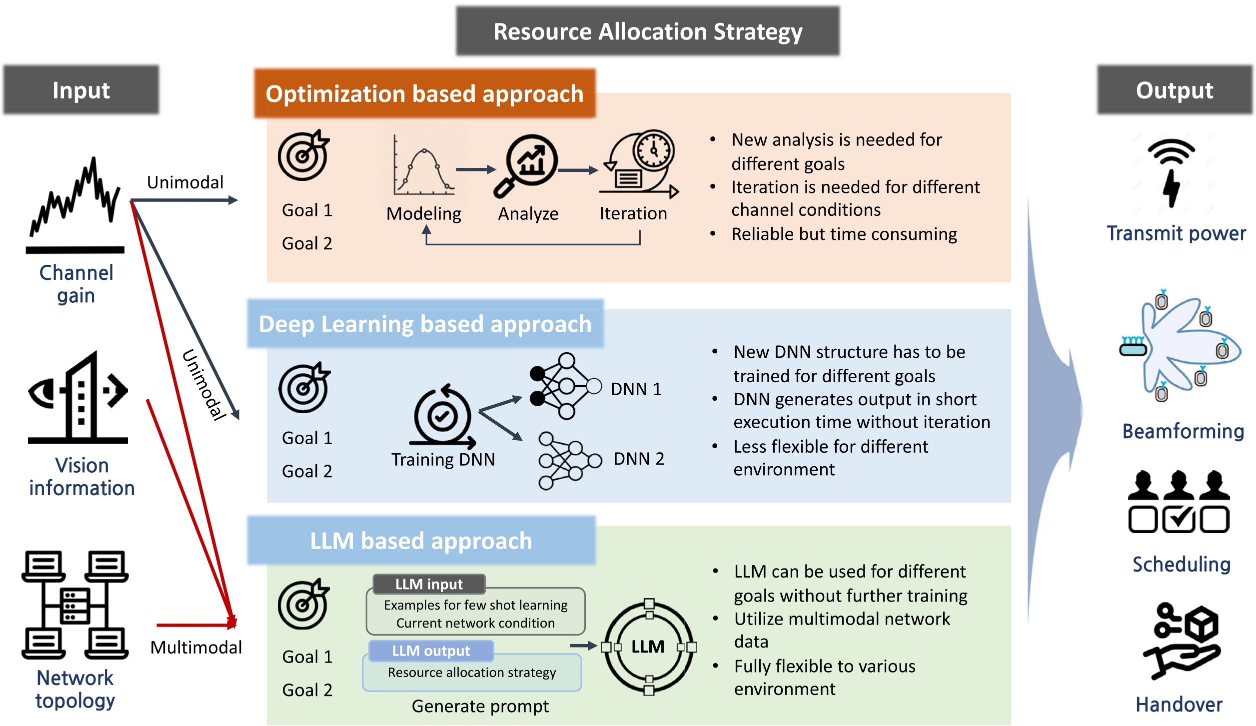

FIGURE 1. Comparison between different approaches for resource allocation.

between an agent and the environment, allowing the model to learn an optimal policy through feedback. Although RL has shown potential in dynamic settings, it typically requires a large number of interactions and careful reward design to ensure stable learning. Despite their algorithmic efficiency, DL-based methods face a fundamental limitation: a lack of adaptability. Once trained, a DNN acts as a static, task-specific function approximator that generalizes poorly to new scenarios such as different network topologies, alternative objective functions, or previously unseen constraints. Accommodating such changes typically necessitates full retraining, which is both time-consuming and computationally intensive.

In summary, although resource allocation has been extensively investigated in the literature, conventional methods face significant limitations, especially with respect to multimodality and flexibility. In particular, resource allocation in future wireless communications systems should be capable of handling multimodal data, such as geographic information. By effectively utilizing such multimodal data, future channel conditions can be predicted, enabling proactive resource allocation, which is crucial for achieving ultra-low latency. In addition, resource allocation should be adaptable to various communication objectives and topologies to accommodate diverse scenarios. Conventional approaches often lack these capabilities. While the DL-based approaches can be trained to flexibly adapt to changing environments, they require specially designed DNNs, and the training process can be time-consuming.

C. RESOURCE ALLOCATION USING LLM

LLM-based resource allocation offers a promising alternative to traditional approaches, marking a paradigm shift from designing task-specific algorithms to leveraging a general-purpose reasoning engine. In this framework, resource allocation decisions can be made by the LLM using a few-shot learning approach, thereby eliminating the need for retraining in each new scenario. Essentially, the LLM serves as a general-purpose optimizer capable of addressing a broad spectrum of resource allocation problems. The key advantages of employing LLMs for resource allocation can be summarized as follows:

- **Multimodality:** An LLM-based scheme is capable of handling diverse multimodal data formats, extending beyond traditional inputs such as channel state information. For example, it can process visual data from surveillance cameras or textual data extracted from network event logs. By effectively integrating such heterogeneous information sources, the system can enable more efficient and proactive resource allocation.
- **Flexibility:** While conventional DL-based methods, such as [39], have demonstrated strong predictive performance for resource management, their applicability remains limited due to task-specific model design and retraining overhead. In contrast, the LLM-based framework provides general-purpose reasoning flexibility, allowing adaptation to new optimization objectives through prompt modification rather than model retraining. For instance, a network operator can shift the system objective from maximizing SE to maximizing

EE while maintaining a minimum user data rate, simply by modifying the text-based prompt provided to the LLM, eliminating the need for costly retraining cycles.

- **Implementation:** As LLMs are expected to become an integral part of future wireless communications systems for various functions (e.g., network monitoring, diagnostics), leveraging this same, pre-existing module for resource allocation simplifies the overall system architecture. Instead of developing and maintaining multiple, disparate optimization algorithms for different tasks, a single, unified LLM can serve as a central control and reasoning engine. This approach reduces development complexity and lowers the barrier for implementing sophisticated, adaptive resource management strategies.

To effectively use LLMs for resource allocation, it is necessary to validate that they can solve mathematical problems. There have been several attempts to use LLMs to solve linguistically described mathematical problems, such as the GSM8K dataset. Although LLMs are primarily designed for linguistic tasks and may not be well suited for rigorous mathematical computations, recent studies indicate that LLMs can indeed solve mathematical problems when properly prompted. For example, the authors of [41] showed that LLMs can accurately solve mathematical problems by formulating prompts, achieving an accuracy of 97.1% in a zero-shot setting. Moreover, in [42], it was confirmed that the mathematical capabilities of LLMs could be enhanced through sequential reasoning techniques such as CoTs, Tree of Thoughts, and Graph of Thoughts. In [43], the use of external math solvers was explored to address the limitations of LLM in precise algebraic computation. In addition, the authors of [44] demonstrated that LLMs could be used to solve optimization problems, such as the linear regression and the traveling salesman problem, through iterative optimization processes. Despite these previous works, there have been no attempts to apply LLMs to resource allocation in wireless communication systems.

In the following sections, we demonstrate the feasibility of employing LLMs for resource allocation. Specifically, Section III introduces a simplified formulation of the resource allocation problem. Section IV then describes how LLMs can be utilized to address this problem and presents a complementary scheme designed to mitigate the limitations of a purely LLM-based approach. Finally, Section V evaluates the feasibility of the proposed LLM-based resource allocation method through simulation results.

III. SYSTEM MODEL AND CONSIDERED RESOURCE ALLOCATION

We consider a two-user interference channel, where each transmitter communicates with its intended receiver over a shared wireless channel. Let the transmitter–receiver pairs be indexed by

$$\mathbb{I} = \{1, 2\}. \tag{1}$$

Each node is equipped with a single antenna. Moreover, the channel gain from transmitter i to receiver j is denoted by $h_{i,j}$, modeled as

$$h_{i,j} = \kappa_{i,j}\sqrt{L_{i,j}}, \tag{2}$$

where $\kappa_{i,j}$ denotes small-scale Rayleigh fading and is modeled as a circularly symmetric complex Gaussian random variable with zero mean and unit variance, i.e., $\kappa_{i,j} \sim \mathcal{CN}(0, 1)$. The large-scale path loss $L_{i,j}$ is expressed as follows:

$$L_{i,j} = G d_{i,j}^{-\gamma}, \tag{3}$$

where γ is the path-loss exponent and G is the path-loss coefficient. We adopt a block-fading model in which all channel coefficients remain constant over one transmission frame and vary independently across frames.

The transmit power of pair i is denoted by P_i, and is constrained by the maximum allowable transmit power. Accordingly, the following condition must hold:

$$0 \leq P_i \leq P_{\mathrm{M}}, \qquad i \in \mathbb{I}, \tag{4}$$

where P_{M} denotes the maximum transmit power.

We note that while the considered system model is somewhat simplified, it captures the essential features of realistic resource allocation problems. First, $h_{i,j}$ properly reflects dynamic wireless environments by incorporating both long-term fading (including path loss) and short-term fading effects. Second, we account for shared wireless channels, which introduce mutual interference, making tractable convex optimization methods such as water filling inapplicable. Although more complex network environments could be analyzed, such exploration is beyond the scope of our current study due to space limitations. The primary objective of this article is to illustrate the principles of LLM and evaluate its feasibility in a simplified toy network setup.

We consider two resource allocation strategies to maximize either total SE or EE by properly adjusting the transmit power. The achievable SE of the transmit pair i, denoted as SE_i, can be expressed as

$$\mathrm{SE}_i = \log_2\left(1 + \frac{h_{i,i}P_i}{N_0 W + \sum_{l\in\{1,2\}\setminus\{i\}} h_{l,i}P_l}\right), \tag{5}$$

where N_0 and W are noise power density and bandwidth, respectively. Then the optimization problem to maximize the total SE, i.e., summation of all SEs can be written as follows:

$$\begin{aligned} &\text{maximize} \quad \sum_{i\in\mathbb{I}} \mathrm{SE}_i \\ &\text{s.t.} \qquad (4). \end{aligned} \tag{6}$$

Conversely, the optimization problem for maximizing total EE can be formulated as:

$$\begin{aligned} &\text{maximize} \quad \sum_{i\in\mathbb{I}} \frac{\mathrm{SE}_i}{P_i + P_{\mathrm{C}}} \\ &\text{s.t.} \qquad (4). \end{aligned} \tag{7}$$

where P_C is the constant circuit power of the transmitter. Although we do not explicitly show it in the paper, the strategy for transmit power control behaves differently depending on whether the goal is to maximize SE or EE. In particular, binary transmit power control has been shown to be optimal for maximizing SE in interference-limited wireless networks [45], [46], [47], whereas maximizing EE generally requires exploring a broader range of transmit power levels rather than strictly binary values. This distinction will later affect the performance of the LLM-based resource allocation.

IV. PROPOSED RESOURCE ALLOCATION BASED ON LLM

In the proposed LLM-based resource allocation strategy, the transmit power of each transmitter is determined based on the channel gain, $h_{i,j}$. In this approach, the channel gain serves as the input, and the transmit power is the output of the LLM, such that the channel gain is provided as a prompt, and the resource allocation can be derived from the text generated by the LLM. Unlike the conventional DNN-based approach, which requires a customized structure, a generic LLM can be used for resource allocation. The procedure of proposed LLM-based resource allocation is depicted in Fig. 2.

In our work, we adopt a few-shot learning-based approach, where the channel gains and the corresponding transmit power strategies are provided as references in the prompt of LLM. Notably, we do not provide a formal description of the optimization objective; instead, we present the channel gains together with their corresponding optimal power allocation, where the optimal performance is obtained via exhaustive search. This approach is computationally feasible in our setting because the network consists of only two transmitter–receiver pairs, making exhaustive enumeration of all power combinations tractable. As a result, our framework can be applied to other optimization problems as well. We also note that these channel gains and transmit power strategies needed for the training can be precomputed or obtained during the execution of the transmit power control.

In the generation of prompts for the LLM, we normalize the channel gain to have zero mean and unit variance, then multiply it by 100. This value is subsequently rounded to an integer. Similarly, the transmit power is normalized by its maximum value, i.e., P_M, and multiplied by 100, and is rounded to an integer. This preprocessing of numerical values is crucial because the LLM recognizes numeric values as strings, making it important to simplify these values for efficiency. Specifically, the original channel gain is typically on the order of 10^{-9}, which is inefficient to express in string format. Additionally, given that the number of inputs to the LLM (i.e., the number of tokens) is limited, it is desirable to reduce the number of characters used to express each numeric value. For example, the formulated prompt to maximize the total SE can be illustrated as follows:

"If A is −29, 30, 128, −26, then B is 0, 100. If A is −31, −31, −19, −31, then B is 100, 0. ⋯ If A is 51, −32, −22, −19, then B is "

In the illustrative example above, the channel gain is denoted as A in the order of $\hat{h}_{1,1}$, $\hat{h}_{1,2}$, $\hat{h}_{2,1}$, $\hat{h}_{2,2}$, where $\hat{h}_{i,j}$ is the preprocessed channel gain $h_{i,j}$. On the other hand, the transmit power aimed at maximizing the total SE, is denoted as B in the order of $\hat{P}_1$, $\hat{P}_2$, where $\hat{P}_i$ represents the preprocessed transmit power. In the considered few-shot learning approach, the training data includes the values of both A and B. However, for the channel gain in the last line, where we need to determine the transmit power, the value of B is omitted. Since the LLM is designed to generate text that follows the provided prompt, it will generate the appropriate resource allocation for the given channel gain. The first output of the LLM is then converted to the transmit power of each transmitter. Note that if the output of the LLM is not in the form $\hat{P}_1$, $\hat{P}_2$, we assume that the LLM fails to find the appropriate transmit power, such that we set the transmit power for both transmitters to zero.

More specifically, unlike conventional DL-based approaches in which transmit powers are directly produced as numeric outputs, the LLM generates decisions in natural-language form. Therefore, the textual output must be parsed and mapped to numeric power levels that satisfy a predefined format and feasible range. If the response does not follow the required structure or yields values outside the valid domain (e.g., two integer power levels between 0 and 100), the power allocation cannot be reliably determined. To address this issue, we incorporate a simple binary transmit power control scheme [48] as a fallback strategy. In this binary scheme, only one transmitter operates at its maximum power while the other remains silent. For the two-user scenario, only two cases need to be evaluated: (i) transmitter 1 transmits at full power, or (ii) transmitter 2 transmits at full power. We compute the achievable rate for both cases and compare them with the rate obtained from the LLM-based allocation. If the LLM output is valid and yields higher performance, it is selected; otherwise, the binary scheme is used as a robust alternative. We emphasize that the LLM-based method is not designed or trained to mimic the binary power control scheme. The binary strategy is used only as a fallback option when the LLM output is incomplete or infeasible.

Regarding the computational complexity during inference, following the Transformer architecture, the inference cost increases approximately quadratically with the number of input tokens due to the self-attention operation, and super-linearly with the model parameter size [30], [49]. Accordingly, the overall inference complexity can be expressed as

$$T_{\text{inf}} = \mathcal{O}\Big(f(N_P) \cdot N_S^2\Big), \tag{8}$$

where N_P denotes the number of model parameters (e.g., 7B or 13B), N_S represents the number of few-shot examples embedded into the prompt, and $f(N_P)$ is a monotonically increasing function that reflects the empirical latency scaling with model size. This formulation captures both the quadratic dependency on the prompt length and the super-linear latency

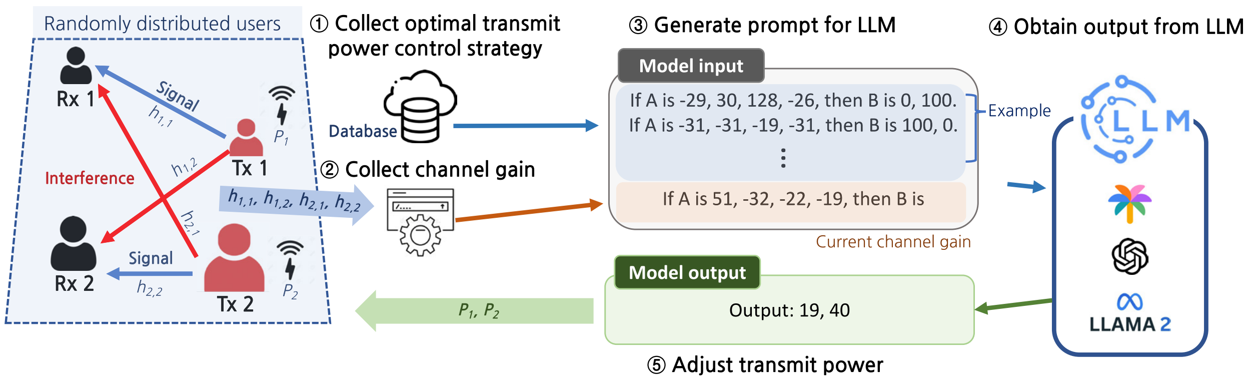

FIGURE 2. Considered system model and procedure for LLM-based resource allocation.

growth observed in recent LLM research. Therefore, the overall computation time tends to increase when a larger LLM base model is employed or when a greater number of few-shot examples are incorporated into the prompt.

V. PERFORMANCE EVALUATION

In the performance evaluation, we assumed that the transmitters and receivers were randomly distributed over an area of 30 m × 30 m. We used the following parameters by default: $W = 10$ MHz, $N_0 = -173$ dBm/Hz, $P_M = 20$ dBm, and $P_C = 30$ dBm. We considered a simplified path loss model with a path loss coefficient of $10^{3.453}$ and a path loss exponent of 3.8. In addition, we used an independent and identically distributed circularly symmetric complex Gaussian random variable for multipath fading, with zero mean and unit variance.

For the LLM structure, we considered three different LLM models based on LLaMA, each fine-tuned on different datasets, namely CodeLLaMA-7B, LLaMA-2-7B-32K-Instruct, and LLaMA-2-7B. First, the CodeLLaMA-7B model, which we refer to as *LLM 1* in our performance evaluation, is fine-tuned using source code, making it specialized for programming tasks and offering robust support for code generation and understanding. Second, the LLaMA-2-7B-32K-Instruct model, denoted as *LLM 2* in our performance evaluation, is fine-tuned to follow complex instructions with a relatively long context length. This model is well-suited for tasks that require detailed execution of instructions. Third, the LLaMA-2-7B model, which we refer to as *LLM 3*, is the base LLaMA 2 model that has not been fine-tuned for specific tasks. All considered models have 7 billion parameters, which are quantized to 5 bits to reduce the model size. The source code utilized for the simulation experiments is publicly available at [51].

In the performance evaluation, we considered two proposed schemes. First, we considered the purely LLM-based scheme, which we refer to as *Prop. 1*. This approach relies solely on LLM for resource allocation. Second, we evaluated a joint scheme that combines the LLM-based approach with binary transmit power control, which we refer to as

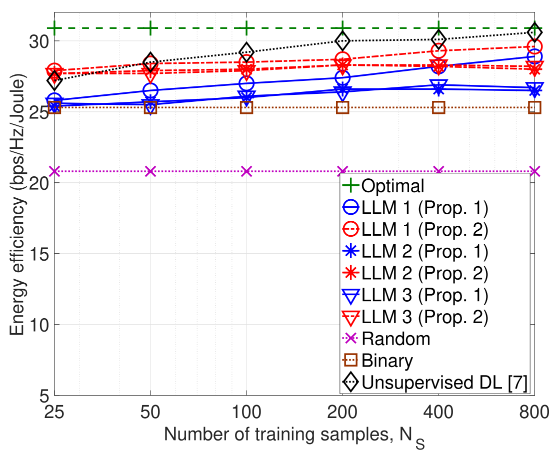

FIGURE 3. EE versus the number of training samples for 7B models.

Prop. 2. In this scheme, the LLM is used together with a simple binary transmit power control method to improve the reliability and mitigate the shortcomings of the purely LLM-based approach. We also present the optimal performance obtained by exhaustive search as a benchmark. Moreover, we include the performance of a random scheme, where the transmit power is randomly assigned, and a binary transmit power control scheme. Furthermore, for comparison, we also consider the unsupervised DL-based resource allocation scheme proposed in [7] and [48], which aims to maximize either SE or EE. This method employs two separate DNN blocks to determine the transmit power and channel selection, where the sigmoid and softmax functions are used at the output layer to ensure that the resource allocation strictly satisfies the given constraints.

First, in Fig. 3, we show the achievable EE by varying the number of training samples fed through the prompts of the LLM, denoted as N_S. As the number of training samples increases, the achievable EE of the LLM-based approaches also increases, achieving up to 96% of the optimal performance in the best case. Notably, even with only 25 training samples, the LLM-based approach outperforms the binary transmit power control scheme. In addition, we can

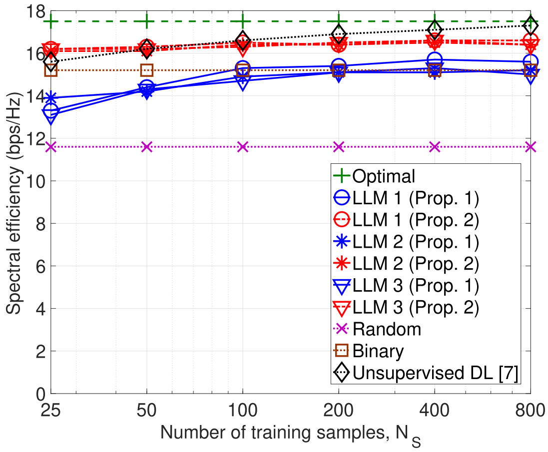

FIGURE 4. SE versus the number of training samples for 7B models.

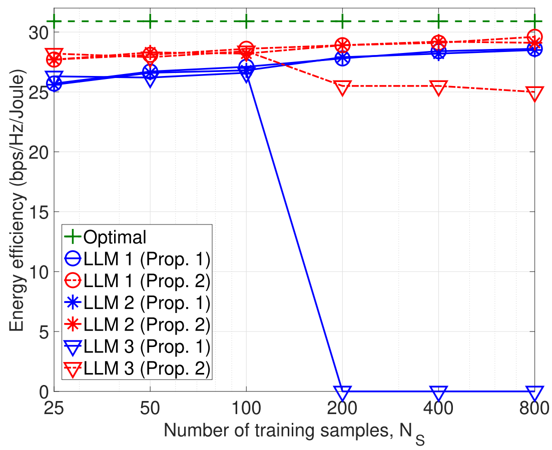

FIGURE 5. EE versus the number of training samples for 13B models.

find that the achievable EE of the LLM-based scheme is significantly higher than that of the random scheme, confirming the importance of proper resource allocation. The unsupervised DL-based approach, whose performance improves as N_S increases, achieves lower performance for small values of N_S, while surpassing the LLM-based approach when sufficient training samples are available. This result confirms the necessity of further optimizing the LLM-based approach. It is worth noting that although the proposed LLM-based approach exhibits slightly lower EE than the unsupervised DL-based approach, its performance remains sufficiently close to the optimal result, and the main contribution of this work lies in validating the feasibility of applying LLMs to resource allocation problems.

By comparing the achievable EE of Prop. 1 (purely LLM-based) and Prop. 2 (joint LLM and binary transmit power control) for the same LLM models, we observe an improvement in EE of about 2-10% in Prop. 2. This result validates the need to integrate conventional resource allocation strategies to improve the performance of LLM-based approaches. Furthermore, the performance varies significantly among the LLM models, with LLM 1 (CodeLLaMA-7B) achieving about 10% higher EE than the others. We analyze that this superiority stems from the strong correlation between the nature of the resource allocation problem and the specialized abilities of CodeLLaMA. The resource allocation task is essentially a structured mapping problem, where numerical inputs (channel gains) are transformed into numerical outputs (transmit powers) based on a set of implicit rules and constraints. This process shares a fundamental similarity with code generation, which requires strict adherence to syntax, logic, and structural patterns. Therefore, CodeLLaMA, being fine-tuned on vast amounts of source code, has developed a superior capability for logical reasoning and pattern recognition within structured domains, making it naturally more adept at solving this type of optimization problem than general-purpose language models.

In Fig. 4, we show the achievable SE as the number of training samples is varied. As expected, the achievable SE increases with the number of training samples, and both SE and EE tend to improve as the number of training samples increases, indicating that larger prompt sizes generally lead to better overall performance. However, compared to EE maximization, the performance gains of the LLM-based approach for SE are relatively small. Specifically, the SE achieved by Prop. 1 (purely LLM-based) is only 3% higher than that of the binary transmit power control scheme, achieving 94.8% of the optimal performance in the best case. We analyze that this is rooted in the fundamental difference between the two optimization objectives from an approximation perspective. EE maximization involves fine-tuning power levels across a continuous range, resembling a quasi-continuous function approximation problem where LLMs excel at capturing subtle and complex patterns. In contrast, the optimal strategy for SE maximization often approaches a discrete on/off decision, akin to a step-function. LLMs can face challenges in accurately approximating such discontinuous functions with abrupt changes, which explains the more limited performance gain over the simple binary scheme. This finding provides valuable insight into the types of optimization problems where LLMs are likely to be most effective. Moreover, similar to the results in Fig. 3, LLM 1 (CodeLLaMA-7B) outperforms the other LLM models. Furthermore, the joint utilization of conventional schemes alongside the LLM-based approach, i.e., Prop. 2, improves the achievable SE by 5-10%. In addition, similar to the previous simulation results, we observe that the unsupervised DL-based approach yields lower SE when N_S is small, while it surpasses the LLM-based approach for larger values of N_S. This again validates the necessity of further improving the LLM-based approach.

In Fig. 5, we show the achievable EE as the number of training samples is varied for larger LLM models, specifically those with 13 billion parameters. For clarity, we omit the performance data of the three baseline schemes, namely the DL-based scheme, the random scheme, and the binary

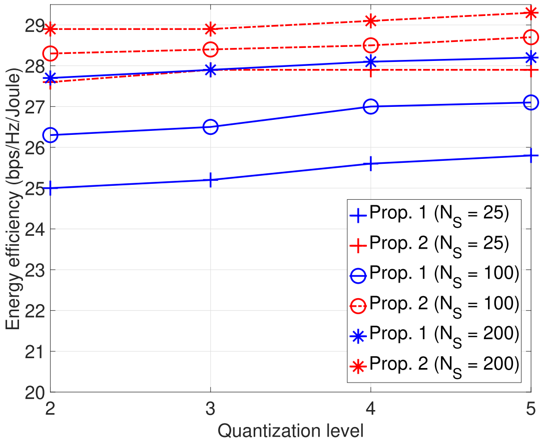

FIGURE 6. **EE versus quantization level for 7B models.**

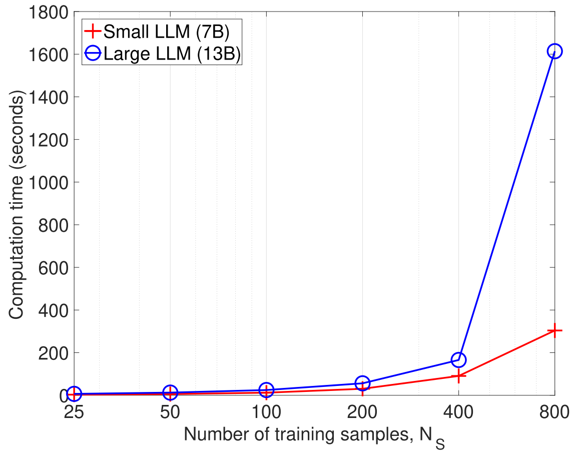

FIGURE 7. **Computation time versus the number of few-shot samples N_S for different model sizes.**

transmit power control. We observe that the overall trend is similar to that in Fig. 3, with EE improving as the number of training samples increases. However, the performance improvement gained from using larger LLM models is relatively small, with an increase of less than 1%. This suggests that the use of large LLM models may not be necessary for resource allocation tasks, which is beneficial for communication systems where the use of large LLMs is challenging. Notably, we find that LLM 3 (the base LLaMA 2) fails to generate a viable resource allocation strategy when the number of training samples exceeds 100. During the experiment, we discovered that the LLM 3 begins to generate output in a different format in these cases, highlighting the importance of integrating conventional schemes that can compensate for the potential malfunctions or inconsistencies of LLM-based approaches.

In Fig. 6, we compare the performance of the LLM-based approach across different quantization levels of the LLM. It is important to note that quantizing parameters in LLM models is a common practice due to the considerable size of unquantized models. For simplicity, only LLM 1 (CodeLLaMA-7B) was considered in the simulation. As expected, the achievable EE improves with higher quantization levels, although the improvement is relatively small. In particular, the performance improvement is more significant when the number of training samples is small, showing a 3.2% improvement in EE. This improvement decreases as the number of training samples increases, with a 1.8% improvement observed when the sample size reaches 200. This result suggests that smaller LLMs can effectively perform resource allocation tasks, highlighting their potential for deployment in practical wireless communication systems with limited computational resources.

Finally, Fig. 7 shows the computation time of the proposed scheme as the number of training samples, N_S, varies for different LLM sizes. For the small LLM, we consider base models with 7 billion parameters, whereas the large LLM corresponds to models with 13 billion parameters. Since different pretrained versions (e.g., CodeLLaMA-7B, LLaMA-2-7B-32K-Instruct, and LLaMA-2-7B) exhibit similar latency, we do not differentiate among the pretraining schemes. As expected, the required computation time increases with N_S because of the ICL process, where the training samples are embedded into the prompt. When N_S is small, the computation time is on the order of only a few seconds; however, it grows beyond 1000 seconds as N_S increases, which is prohibitively high for real-time resource allocation. The larger LLM consistently requires longer computation time, primarily due to limited computational resources in our current hardware setting, and this bottleneck becomes more pronounced for longer prompts owing to memory constraints. These results highlight one of the main obstacles to applying LLM-based approaches, namely the heavy computational cost and processing capability requirements. Such limitations may be mitigated by leveraging cloud-based computation or employing smaller, distilled LLMs specialized for wireless communication tasks.

VI. RESEARCH CHALLENGES

This section discusses the main challenges and current limitations of applying LLMs to wireless resource allocation, providing insights into the factors that constrain their scalability, reliability, and real-time applicability.

A. LATENCY AND COMPUTATION TIME

The size of LLMs is generally a significant factor due to their large number of parameters; GPT-4, for example, has 1.7 trillion parameters. As confirmed by our simulation results, even relatively small LLMs with 7 or 13 billion parameters require substantial computation time. Since practical power allocation decisions often need to be made within millisecond-level time scales, this discrepancy highlights a critical barrier to real-time deployment. Moreover, the large computational footprint of LLMs frequently necessitates cloud-based processing, which further increases latency due to communication and queuing delays. To address these challenges, the development of small LLMs (sLLMs) tailored for resource allocation is essential. These sLLMs, combined with edge AI technology, can distribute the computational load across distributed nodes, thereby reducing latency and improving efficiency. Going a step further, a more promising direction involves a hierarchical LLM architecture. In this approach, a highly efficient sLLM deployed at the network edge handles the majority of standard resource allocation scenarios instantly. For the remaining complex or ambiguous cases, the sLLM escalates the problem to a powerful, cloud-based LLM for more thorough analysis. This hybrid computational model effectively balances the trade-off between latency and optimality, thereby enhancing the practicality of LLM-based resource allocation.

B. OPTIMIZED LLM ARCHITECTURE

To achieve higher performance and practical deployment, the model architecture itself must be optimized to meet the specific demands of wireless networks. Relying solely on generic pre-trained models is insufficient. Future research should explore the design of specialized sLLMs tailored for network optimization tasks. Techniques such as quantization and knowledge distillation offer promising pathways to building compact yet powerful models. Furthermore, fully leveraging the potential of LLMs requires enriching the input space beyond conventional channel state information. Incorporating diverse multimodal data, such as real-time user mobility patterns obtained via GPS or predicted network traffic load, can facilitate proactive and context-aware resource allocation strategies. Lastly, the energy consumption of LLM inference remains a critical concern, particularly for deployment in latency- and power-constrained wireless environments. As model size increases, so does the computational burden, potentially offsetting performance gains. To address this, energy-aware model design—through techniques such as pruning, quantization, or adaptive inference—should become a key focus, enabling scalable and sustainable use of LLMs in real-world networks. Moreover, beyond architectural and energy optimizations, improving data efficiency under limited or imperfect supervision also represents an important future direction. Recent studies, such as [52] and [53], have shown that contrastive self-supervised and meta-learning techniques can enhance robustness in signal-level tasks with unlabeled or noisy data. Incorporating similar strategies into LLM-based resource allocation could enable more reliable and adaptive operation in realistic wireless environments where perfect supervision is unavailable.

C. TRAINING METHODOLOGY

While the proposed few-shot learning approach demonstrates the feasibility of using LLMs for resource allocation, its reliance on pre-calculated optimal labels poses scalability challenges, as generating such labeled data is computationally expensive and often impractical for large-scale or dynamically changing networks. To address this limitation, future research should explore more autonomous and self-adaptive learning methodologies. One promising direction is RL, enabling the LLM to operate as an intelligent agent that interacts directly with a simulated network environment to learn optimal strategies without labeled data. Such an approach could enhance adaptability to unseen conditions while reducing the dependence on offline optimization. In parallel, advancements in prompt engineering are essential to improve scalability and efficiency by enabling the model to handle larger user populations, more complex optimization objectives, and dynamic constraints within a limited token context. Furthermore, efficient fine-tuning techniques such as Low-Rank Adaptation (LoRA) should be investigated to adapt pretrained LLMs to specific wireless environments with minimal computational overhead. Finally, incorporating continuous learning principles into LLM-based resource allocation represents another promising direction. Recent advances, such as [54], demonstrate that continual adaptation mechanisms can mitigate catastrophic forgetting and enhance long-term stability in dynamically evolving wireless environments.

D. INTERPRETABILITY AND EXPLAINABILITY

Similar to DL-based approaches, the "closed-box" nature of LLMs poses significant challenges for their adoption in mission-critical communication systems. The inherent stochasticity of LLM-generated outputs can lead to variations in decisions and potential instability, while hallucinated or infeasible outputs may cause unsafe network behavior. Although our hybrid design which combines the LLM with a low-complexity binary scheme, partially alleviates this issue, further improvements are needed. Future work should incorporate principles from Explainable AI (XAI) and controllable inference design to enhance reliability and interpretability. By prompting the LLM to generate both decisions and their underlying rationales, its reasoning process can be made more transparent. Additionally, adopting structured output formats would enable automated verification and constraint checking through rule-based post-processing. Together with the proposed hybrid architecture, these approaches can improve stability, reproducibility, and trustworthiness in LLM-based wireless resource management.

VII. CONCLUSION

In this article, we pioneered a novel approach for wireless resource allocation, shifting the paradigm from rigid, task-specific conventional DL to a flexible, prompt-driven framework powered by LLMs. By grounding our investigation in a representative resource allocation problem, we demonstrated the inherent capability of LLMs to interpret system states and generate effective resource allocation strategies without the need for task-specific model design or extensive retraining. Recognizing the current limitations in the reliability of LLMs, we proposed a pragmatic hybrid strategy that integrates a conventional allocation method as a fallback mechanism, improving both system performance and deployment feasibility. Our performance evaluation confirmed that this

LLM-based approach could achieve near-optimal performance, and our analysis revealed key characteristics, such as the superior performance of code-specialized models and the direct impact of few-shot example quality. Finally, by outlining critical research challenges such as latency and explainability, this work provided a concrete roadmap for maturing LLMs from a novel concept into practical and robust solvers for dynamic resource allocation in wireless communications systems.

An important direction for future work is to extend the proposed framework to more realistic and complex propagation environments. In particular, incorporating advanced channel generation systems, such as geometry-based stochastic or ray-tracing-based models, would enable a more accurate validation of the LLM's capability under multi-path and spatially correlated channels. Furthermore, extending the evaluation to multi-user and multi-cell scenarios, together with RL-based autonomous training, will further enhance scalability and adaptability. Finally, exploring lightweight fine-tuning and prompt optimization techniques can improve computational efficiency and facilitate real-time deployment in practical wireless networks.

REFERENCES

[1] Y. Xu, G. Gui, H. Gacanin, and F. Adachi, "A survey on resource allocation for 5G heterogeneous networks: Current research, future trends, and challenges," *IEEE Commun. Surveys Tuts.*, vol. 23, no. 2, pp. 668–695, 2nd Quart., 2021.

[2] H. Liang and W. Zhang, "A survey and taxonomy of resource allocation methods in wireless networks," *J. Commun. Inf. Netw.*, vol. 6, no. 4, pp. 372–384, Dec. 2021.

[3] H. Tataria, M. Shafi, A. F. Molisch, M. Dohler, H. Sjöland, and F. Tufvesson, "6G wireless systems: Vision, requirements, challenges, insights, and opportunities," *Proc. IEEE*, vol. 109, no. 7, pp. 1166–1199, Jul. 2021.

[4] Y.-F. Liu, T.-H. Chang, M. Hong, Z. Wu, A. Man-Cho So, E. A. Jorswieck, and W. Yu, "A survey of recent advances in optimization methods for wireless communications," *IEEE J. Sel. Areas Commun.*, vol. 42, no. 11, pp. 2992–3031, Nov. 2024.

[5] R. Chen, F. Shi, and C. Zhao, "A novel energy-efficient power allocation for D2D communications underlaying cellular network," in *Proc. 11th Int. Conf. Wireless Commun. Signal Process. (WCSP)*, Oct. 2019, pp. 1–6.

[6] H. Wang, J. Wang, G. Ding, L. Wang, T. A. Tsiftsis, and P. K. Sharma, "Resource allocation for energy harvesting-powered D2D communication underlaying UAV-assisted networks," *IEEE Trans. Green Commun. Netw.*, vol. 2, no. 1, pp. 14–24, Mar. 2018.

[7] W. Lee, O. Jo, and M. Kim, "Intelligent resource allocation in wireless communications systems," *IEEE Commun. Mag.*, vol. 58, no. 1, pp. 100–105, Jan. 2020.

[8] Y. Yuan, T. Yang, H. Feng, and B. Hu, "An iterative matching-stackelberg game model for channel-power allocation in D2D underlaid cellular networks," *IEEE Trans. Wireless Commun.*, vol. 17, no. 11, pp. 7456–7471, Nov. 2018.

[9] S. Dominic and L. Jacob, "Distributed resource allocation for D2D communications underlaying cellular networks in time-varying environment," *IEEE Commun. Lett.*, vol. 22, no. 2, pp. 388–391, Feb. 2018.

[10] C. Kim, H.-H. Choi, and K. Lee, "Joint optimization of trajectory and resource allocation for multi-UAV-enabled wireless-powered communication networks," *IEEE Trans. Commun.*, vol. 72, no. 9, pp. 5752–5764, Sep. 2024.

[11] H. Sun, X. Chen, Q. Shi, M. Hong, X. Fu, and N. D. Sidiropoulos, "Learning to optimize: Training deep neural networks for interference management," *IEEE Trans. Signal Process.*, vol. 66, no. 20, pp. 5438–5453, Oct. 2018.

[12] F. Liang, C. Shen, W. Yu, and F. Wu, "Towards optimal power control via ensembling deep neural networks," *IEEE Trans. Commun.*, vol. 68, no. 3, pp. 1760–1776, Mar. 2020.

[13] P. de Kerret, D. Gesbert, and M. Filippone, "Team deep neural networks for interference channels," in *Proc. IEEE Int. Conf. Commun. Workshops (ICC Workshops)*, Kansas City, MO, USA, May 2018, pp. 1–6.

[14] Y. Shi, L. Lian, Y. Shi, Z. Wang, Y. Zhou, L. Fu, L. Bai, J. Zhang, and W. Zhang, "Machine learning for large-scale optimization in 6G wireless networks," *IEEE Commun. Surveys Tuts.*, vol. 25, no. 4, pp. 2088–2132, 4th Quart., 2023.

[15] W. Lee, M. Kim, and D.-H. Cho, "Transmit power control using deep neural network for underlay device-to-device communication," *IEEE Wireless Commun. Lett.*, vol. 8, no. 1, pp. 141–144, Feb. 2019.

[16] W. Lee, K. Lee, and T. Q. S. Quek, "Deep-learning-assisted wireless-powered secure communications with imperfect channel state information," *IEEE Internet Things J.*, vol. 9, no. 13, pp. 11464–11476, Jul. 2022.

[17] W. Lee and R. Schober, "Deep learning-based resource allocation for device-to-device communication," *IEEE Trans. Wireless Commun.*, vol. 21, no. 7, pp. 5235–5250, Jul. 2022.

[18] K. Heo, W. Lee, and K. Lee, "UAV-assisted wireless-powered secure communications: Integration of optimization and deep learning," *IEEE Trans. Wireless Commun.*, vol. 23, no. 9, pp. 10530–10545, Sep. 2024.

[19] H. Lee, S. H. Lee, and T. Q. S. Quek, "Deep learning for distributed optimization: Applications to wireless resource management," *IEEE J. Sel. Areas Commun.*, vol. 37, no. 10, pp. 2251–2266, Oct. 2019.

[20] Z. Wang, M. Eisen, and A. Ribeiro, "Learning decentralized wireless resource allocations with graph neural networks," *IEEE Trans. Signal Process.*, vol. 70, pp. 1850–1863, 2022.

[21] L. Bariah, Q. Zhao, H. Zou, Y. Tian, F. Bader, and M. Debbah, "Large generative AI models for telecom: The next big thing?" *IEEE Commun. Mag.*, vol. 62, no. 11, pp. 84–90, Nov. 2024.

[22] X. Wang, J. Zhu, R. Zhang, L. Feng, D. Niyato, J. Wang, H. Du, S. Mao, and Z. Han, "Chain-of-thought for large language model-empowered wireless communications," 2025, *arXiv:2505.22320*.

[23] H. Zhou, C. Hu, D. Yuan, Y. Yuan, D. Wu, X. Chen, H. Tabassum, and X. Liu, "Large language models for wireless networks: An overview from the prompt engineering perspective," *IEEE Wireless Commun.*, vol. 32, no. 4, pp. 98–106, Aug. 2025.

[24] H. Zhou, C. Hu, Y. Yuan, Y. Cui, Y. Jin, C. Chen, H. Wu, D. Yuan, L. Jiang, D. Wu, X. Liu, J. Zhang, X. Wang, and J. Liu, "Large language model (LLM) for telecommunications: A comprehensive survey on principles, key techniques, and opportunities," *IEEE Commun. Surveys Tuts.*, vol. 27, no. 3, pp. 1955–2005, Jun. 2025.

[25] J. Ahn, R. Verma, R. Lou, D. Liu, R. Zhang, and W. Yin, "Large language models for mathematical reasoning: Progresses and challenges," in *Proc. 18th Conf. Eur. Chapter Assoc. Comput. Linguistics, Student Res. Workshop*, Malta, Mar. 2024, pp. 225–237.

[26] S. Imani, L. Du, and H. Shrivastava, "MathPrompter: Mathematical reasoning using large language models," 2023, *arXiv:2303.05398*.

[27] W. Liu, H. Hu, J. Zhou, Y. Ding, J. Li, J. Zeng, M. He, Q. Chen, B. Jiang, A. Zhou, and L. He, "Mathematical language models: A survey," 2023, *arXiv:2312.07622*.

[28] L. Yu, W. Jiang, H. Shi, J. Yu, Z. Liu, Y. Zhang, J. T. Kwok, Z. Li, A. Weller, and W. Liu, "MetaMath: Bootstrap your own mathematical questions for large language models," in *Proc. 12th Int. Conf. Learn. Represent. (ICLR)*, Vienna, Austria, 2023.

[29] X. Hao, S. Yang, R. Liu, Z. Feng, T. Peng, and B. Huang, "VSLM: Virtual signal large model for few-shot wideband signal detection and recognition," *IEEE Trans. Wireless Commun.*, vol. 24, no. 2, pp. 909–925, Feb. 2025.

[30] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin, "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 30, Long Beach, CA, USA, Dec. 2025, pp. 5998–6008.

[31] Z. Niu, G. Zhong, and H. Yu, "A review on the attention mechanism of deep learning," *Neurocomputing*, vol. 452, pp. 48–62, Sep. 2021.

[32] T. B. Brown, "Language models are few-shot learners," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 33, 2020, pp. 1877–1901.

[33] J. Lee, X. Wang, D. Schuurmans, M. Bosma, E. H., Q. V. Le, D. Zhou, L. Quoc, and Z. Denny, "Chain-of-thought prompting elicits reasoning in large language models," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2022, pp. 24824–24837.

[34] T. Kojima, S. Gu, M. Reid, Y. Matsuo, and Y. Iwasawa, "Large language models are zero-shot reasoners," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2022, pp. 22199–22213.
[35] S. Minaee, T. Mikolov, N. Nikzad, M. Chenaghlu, R. Socher, X. Amatriain, and J. Gao, "Large language models: A survey," 2024, *arXiv:2402.06196*.
[36] A. Grattafiori et al., "The llama 3 herd of models," 2024, *arXiv:2407.21783*.
[37] A. Q. Jiang et al., "Mixtral of experts," 2024, *arXiv:2401.04088*.
[38] A. Liu et al., "DeepSeek-v2: A strong, economical, and efficient mixture-of-experts language model," 2024, *arXiv:2405.04434*.
[39] J. Jalali, J. Roa, Y. Song, R. Zhao, and B. Sheen, "Fast best beam prediction and overhead reduction for 6G networks: A deep learning approach," in *Proc. IEEE 99th Veh. Technol. Conf. (VTC-Spring)*, Singapore, Jun. 2024, pp. 1–6.
[40] W. Lee and K. Lee, "Deep learning frameworks for solving infeasible optimization problems in vehicular communications," *IEEE Open J. Commun. Soc.*, vol. 5, pp. 3289–3298, 2024.
[41] Q. Zhong, K. Wang, Z. Xu, J. Liu, L. Ding, and B. Du, "Achieving >97% on GSM8K: Deeply understanding the problems makes LLMs better solvers for math word problems," 2024, *arXiv:2404.14963*.
[42] M. Besta, N. Blach, A. Kubicek, R. Gerstenberger, M. Podstawski, L. Gianinazzi, J. Gajda, T. Lehmann, H. Niewiadomski, P. Nyczyk, and T. Hoefler, "Graph of thoughts: Solving elaborate problems with large language models," in *Proc. AAAI Conf. Artif. Intell.*, 2024, vol. 38, no. 16, pp. 17682–17690.
[43] J. He-Yueya, G. Poesia, R. E. Wang, and N. D. Goodman, "Solving math word problems by combining language models with symbolic solvers," 2023, *arXiv:2304.09102*.
[44] C. Yang, X. Wang, Y. Lu, H. Liu, Q. V. Le, D. Zhou, and X. Chen, "Large language models as optimizers," 2023, *arXiv:2309.03409*.
[45] H. Inaltekin and S. V. Hanly, "Optimality of binary power control for the single cell uplink," *IEEE Trans. Inf. Theory*, vol. 58, no. 10, pp. 6484–6498, Oct. 2012.
[46] J. M. B. da Silva and G. Fodor, "A binary power control scheme for D2D communications," *IEEE Wireless Commun. Lett.*, vol. 4, no. 6, pp. 669–672, Dec. 2015.
[47] A. Gjendemsjo, D. Gesbert, G. E. Oien, and S. G. Kiani, "Binary power control for sum rate maximization over multiple interfering links," *IEEE Trans. Wireless Commun.*, vol. 7, no. 8, pp. 3164–3173, Aug. 2008.
[48] W. Lee and K. Lee, "Resource allocation scheme for guarantee of QoS in D2D communications using deep neural network," *IEEE Commun. Lett.*, vol. 25, no. 3, pp. 887–891, Mar. 2021.
[49] T. Dao, D. Y. Fu, S. Ermon, A. Rudra, and C. Ré, "FlashAttention: Fast and memory-efficient exact attention with IO-awareness," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, New Orleans, LA, USA, Nov. 2022, pp. 16344–16359.
[50] W. Lee and B. C. Chung, "Ensemble deep learning based resource allocation for multi-channel underlay cognitive radio system," *ICT Exp.*, vol. 9, no. 4, pp. 642–647, Aug. 2023.
[51] W. Lee and J. Park. (2025). *Source Code*. Accessed: Jul. 21, 2025. [Online]. Available: https://github.com/seotaijiya/LLMOPT
[52] X. Hao, Z. Feng, R. Liu, S. Yang, L. Jiao, and R. Luo, "Contrastive self-supervised clustering for specific emitter identification," *IEEE Internet Things J.*, vol. 10, no. 23, pp. 20803–20818, Dec. 2023.
[53] X. Hao, Z. Feng, T. Peng, and S. Yang, "Meta-learning guided label noise distillation for robust signal modulation classification," *IEEE Internet Things J.*, vol. 12, no. 1, pp. 402–418, Jan. 2025.
[54] X. Hao, S. Yang, R. Liu, Z. Feng, T. Peng, and B. Huang, "SMTC-CL: Continuous learning via selective multi-task coordination for adaptive signal classification," *IEEE Trans. Cognit. Commun. Netw.*, vol. 11, no. 3, pp. 1664–1681, Jun. 2025.

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